

The Ontic Derivation Chain:

Twenty-Six Physical Constants from Two Inputs

Abstract

From two inputs—the spatial dimension $D = 3$ and the lemniscate constant $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$ —a chain of algebraic identities yields twenty-six testable physical quantities (some sharing common roots). The chain proceeds: $\varpi \rightarrow M \rightarrow G^* \rightarrow \pi$, where a master quadratic $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$ produces roots $x_+ = 137.036$ and $x_- = 3.024$, identified with $1/\alpha$ (1.26 ppm tree-level; <1 ppt corrected) and $N_c = 3$. An integer cascade yields $\{3, 4, 7, 13, 47\}$, from which follow $\sin^2\theta_W = 3/13$, $\alpha_s(M_Z) = 7/59$, integer lepton mass ratios, the full CKM and PMNS mixing matrices, and the gravitational hierarchy $\alpha_G \propto \alpha^{20}$. A companion proof suite (301/301 checks, NumPy/SciPy) verifies every numerical claim. We present the framework as a self-consistent algebraic structure and state explicit falsification criteria. Companion papers trace the lifecycle of a hydrogen atom through these constants [11], analyze the discrete–continuous bridge in pure mathematics [10], and propose a unifying ontological constructor [12].

Epistemic tags: **[Axiom]** structural postulate; **[Theorem]** proven from axioms; **[Selection]** argued, not uniquely proven; **[Conjecture]** requires experimental validation.

1. Inputs and Notation

The entire derivation chain rests on two inputs.

Input 1 [Axiom]: $D = 3$ spatial dimensions (cubic lattice $\mathbb{Z}^3$).

Input 2 [Axiom]: The lemniscate constant $\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}} = 2.62206\dots$

Table 1 defines the principal symbols.

2. The Derivation Chain (Layers -1 to 3)

2.1 Layers 0 to 1: From e to ϖ

Layer 0. Starting from e (the natural base, unique eigenvalue of d/dx), the Euler–Mascheroni constant $\gamma = \lim_{n \rightarrow \infty} [\sum_{k=1}^n 1/k - \ln n]$ enters the Weierstrass product $1/\Gamma(z) = z e^{\gamma z} \prod_{n \geq 1} [(1 + z/n) e^{-z/n}]$, evaluated at $z = 1/4$ to give $\Gamma(1/4) = 3.62561\dots$ **[Theorem]**

Definition 2.1 (Lemniscate constant). $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62206\dots$

Definition 2.2 (Gauss constant). $M = \varpi/\pi = 1/\text{AGM}(1, \sqrt{2}) = 0.83463\dots$

Symbol	Definition	Value	Layer
ϖ	$\Gamma(1/4)^2/(2\sqrt{2\pi})$	2.62206	1
M	$\varpi/\pi = 1/\text{AGM}(1, \sqrt{2})$	0.83463	1
G^*	$2\sqrt{\varpi \cdot M} = \sqrt{2}\Gamma(1/4)^2/(2\pi)$	2.95868	2
k	Lattice DoF coefficient	16	3
x_+	Larger root of master quadratic	137.036	3
x_-	Smaller root	3.024	3
α	$1/x_+$	7.297×10^{-3}	5
N_c	$\lfloor x_- \rfloor$ (color charges)	3	4
b_3	$(11N_c - 2N_f)/3$ (QCD beta)	7	4
N_{eff}	$b_3 + 2N_c$ (effective DoF)	13	4
$\mathcal{D}$	$N_c \cdot N_{\text{base}}^2 - 1$	47	4

Table 1: Principal symbols of the ontic chain.

2.2 Layer 2: The Universal Operator G^*

Definition 2.3 (G^*). $G^* = 2\sqrt{\varpi \cdot M}$

$$G^* = \frac{\Gamma(1/4)^2}{\sqrt{2}\Gamma(1/2)^2} = \frac{2\varpi}{\sqrt{\pi}} = 2.95868 \dots \quad (1)$$

The first form is π -free: both $\Gamma(1/4)$ and $\Gamma(1/2)$ are definable by integrals over e^{-t} and rational powers of t . The identity $\Gamma(1/2)^2 = \pi$ is a consequence, not an input.

Corollary 2.1 (Derived π). $\pi = 4\varpi^2/G^{*2}$. Since both ϖ (lemniscate integral) and G^* (Γ -ratio) are computable without π , this derivation is not circular. [\[Theorem\]](#)

2.3 Layer 2b: Generalized Discriminant

Lemma 2.2 (Three regimes). The generalized quadratic $x^2 - k G^{*2} x + k G^{*3} = 0$ has discriminant $\Delta = k G^{*3}(k G^* - 4)$. Three domains:

1. $k G^* > 4$ ($k = 16$): $\Delta > 0$, real roots $\rightarrow$ **physics**.
2. $k G^* = 4$ ($k = 4/G^*$): $\Delta = 0$, degenerate $\rightarrow$ Born rule boundary.
3. $k G^* < 4$ ($k = 1/2$): $\Delta < 0$, complex conjugate roots (no physical interpretation in this paper).

2.4 Layer 3: The Master Quadratic

Lemma 2.3 (Coefficient 16). On the minimal $2 \times 2 \times 2$ cubic lattice: 24 total components – 7 gauge constraints (Gauss law) – 1 global phase = 16 physical degrees of freedom. Equivalently, $16 = N_{\text{base}}^2 = 2^{D+1}$. [\[Selection\]](#)

Lemma 2.4 (CM selection). The cubic lattice $\mathbb{Z}^3$ has $\mathbb{Z}_4$ rotation symmetry, selecting the unique elliptic curve with Complex Multiplication discriminant $j = 1728$: the lemniscatic curve $y^2 = x^3 - x$ at modulus $k = 1/\sqrt{2}$. This fixes G^* as the relevant elliptic constant. [\[Selection\]](#)

Remark 2.1 (Why this quadratic). The form $x^2 - kc^2 x + kc^3 = 0$ is selected by requiring the Vieta product-to-sum ratio to equal c itself: $x_+ x_- / (x_+ + x_-) = kc^3 / (kc^2) = c = G^*$. That is, G^* is half the harmonic mean of the two physics roots. The asymmetry between c^2 (sum) and c^3 (product) encodes the distinction between spatial amplitude and spacetime action per degree of freedom. [\[Selection\]](#)

Theorem 2.5 (Master Quadratic).

$$\boxed{x^2 - 16 G^{*2} x + 16 G^{*3} = 0} \quad (2)$$

with roots

$$x_+ = 8G^{*2} \left(1 + \sqrt{1 - \frac{1}{4G^*}} \right) = 137.0362 \dots \quad (3)$$

$$x_- = 8G^{*2} \left(1 - \sqrt{1 - \frac{1}{4G^*}} \right) = 3.0240 \dots \quad (4)$$

Vieta's relations: $x_+ + x_- = 16 G^{*2}$, $x_+ \cdot x_- = 16 G^{*3}$.

Remark 2.2. The harmonic mean identity $G^* = x_+ x_- / (x_+ + x_-)$ shows G^* is half the harmonic mean of the two physics roots.

3. The Integer Cascade (Layer 4)

Proposition 3.1 (Framework integers). *From $x_- = 3.024 \dots$ and $D = 3$, the following integers are fixed:*

Symbol	Formula	Value	Physics	Tag
N_c	$\lfloor x_- \rfloor$	3	Color charges	[Conjecture]
N_{gen}	N_c	3	Fermion generations	[Conjecture]
N_f	$2N_{\text{gen}}$	6	Quark flavors	[Theorem]
N_{base}	$2^{(D+1)/2}$	4	Spinor dimension	[Theorem]
b_3	$(11N_c - 2N_f)/3$	7	QCD beta coefficient	[Theorem]
N_{eff}	$b_3 + 2N_c$	13	Effective DoF = F_7	[Theorem]
$\mathcal{D}$	$N_c N_{\text{base}}^2 - 1$	47	Constraint dimension	[Theorem]

Remark 3.1. $N_{\text{eff}} = 13 = F_7$ (seventh Fibonacci number). $b_3 = N_{\text{base}} + N_c$ (additive closure of the two primitive integers).

3.1 Gauge Groups from the Moore Neighborhood

The 26 neighbors of a voxel in $\mathbb{Z}^3$ decompose uniquely by distance into three sublattices. [\[Theorem\]](#) A neighbor at displacement $(\Delta x, \Delta y, \Delta z)$ couples to the flux components corresponding to its nonzero coordinates:

Sublattice	Count	Distance	J -components	Gauge group
SC (face)	6	1	1	U(1)
FCC (edge)	12	$\sqrt{2}$	2	SU(2)
BCC (corner)	8	$\sqrt{3}$	3	SU(3)

The Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ is the unique orthogonal decomposition of $|\mathbf{J}|^2 = J_x^2 + J_y^2 + J_z^2$ by the Moore sublattices. It is not postulated. It is counted.

4. Coupling Constants (Layer 5)

Four coupling constants follow from the integers and x_+ .

$$\alpha = \frac{1}{x_+} = 7.297 \times 10^{-3} \quad \text{[Conjecture]: tree-level, 1.26 ppm from CODATA} \quad (5)$$

$$\sin^2 \theta_W = \frac{N_c}{N_{\text{eff}}} = \frac{3}{13} = 0.23077 \quad \text{[Conjecture]: 0.19\% from 0.23122} \quad (6)$$

$$\alpha_s(M_Z) = \frac{b_3}{b_3 + 4N_{\text{eff}}} = \frac{7}{59} = 0.11864 \quad \text{[Conjecture]: 0.63\% from 0.1179} \quad (7)$$

$$\alpha_G = 2\pi \left(\frac{16}{3} \right)^2 \left(N_{\text{eff}} + \frac{3}{b_3} \right)^2 \alpha^{20} \quad \text{[Conjecture]: exponent } 20 = N_{\text{eff}} + b_3 \quad (8)$$

The gravitational hierarchy $\alpha_G \approx 5.91 \times 10^{-39}$ arises from α^{20} , explaining the 37-order-of-magnitude gap between electromagnetism and gravity as a *cross-domain* power of the coupling.

5. Mass Hierarchy (Layer 6)

5.1 Absolute Scale

In lattice natural units ($K_B = 1$, $a = 1$, $\tau = 1$), the electron mass *is* the energy unit. The Planck mass is then derived, not imposed:

$$M_P = \frac{K_B}{\sqrt{2\pi} \frac{N_{\text{base}}^2}{N_c} \alpha^{11}} = \frac{1}{\sqrt{2\pi} \frac{16}{3} \alpha^{11}} \approx 2.39 \times 10^{22} \text{ lattice units.} \quad (9)$$

Converting to SI requires one unit convention: $K_B = 0.5100 \text{ MeV}$ (the electron mass). This gives $M_P = 2.39 \times 10^{22} \times 0.5100 \text{ MeV} = 1.22 \times 10^{19} \text{ GeV}$, matching the known Planck mass. **[Selection]**

Equivalently, the electron mass formula reads:

$$m_e = M_P \sqrt{2\pi} \frac{16}{3} \alpha^{11} = 0.5100 \text{ MeV} \quad (0.19\% \text{ from experimental } 0.51100 \text{ MeV}). \quad (10)$$

[Selection]

5.2 Integer Mass Ratios

Ratio	Formula	FTD	Expt.	Err.
m_μ/m_e	$3b_3(b_3 + N_c) - N_c$	207	206.768	0.11%
m_τ/m_e	$(N_{\text{eff}} + N_{\text{base}}) \cdot 207 - 2N_c b_3$	3477	3477.23	0.007%
m_p/m_e	$N_{\text{eff}} \cdot x_+ + T_{10}$	1836.47	1836.15	0.018%

where $T_{10} = (b_3 + N_c)(b_3 + N_c + 1)/2 = 55$ is the 10th triangular number. The proton mass combines the EM contribution ($N_{\text{eff}} \cdot x_+$: valence quarks coupled to the fine structure) with a QCD binding correction (T_{10} , reflecting the $\binom{b_3 + N_c + 1}{2}$ gluon exchange channels). **[Selection]**

5.3 Electroweak Sector

$$v = M_P \sqrt{2\pi} \alpha^8 = 246.08 \text{ GeV} \quad (0.05\% \text{ from } 246.22) \quad \text{[Conjecture]} \quad (11)$$

$$m_H = v \sqrt{\frac{6}{23}} = 125.69 \text{ GeV} \quad (0.35\% \text{ from } 125.25) \quad \text{[Selection]} \quad (12)$$

$$m_W = \frac{e v}{2 \sin \theta_W} = 77.56 \text{ GeV} \quad (3.5\% \text{ from } 80.37) \quad \text{[Parametric Insertion]} \quad (13)$$

$$m_Z = m_W / \cos \theta_W = 88.44 \text{ GeV} \quad (3.0\% \text{ from } 91.19) \quad \text{[Parametric Insertion]} \quad (14)$$

The m_W and m_Z errors ($\sim 3\%$) are the worst mass predictions and propagate from $\sin^2 \theta_W = 3/13 = 0.2308$ vs. the measured 0.23122, combined with the FTD-derived $v = 246.08 \text{ GeV}$. An alternative integer-ratio formula $m_W = [b_3(b_3 + N_c) - N_c] / (8\alpha^2) \cdot m_e = 80.21 \text{ GeV}$ achieves 0.2% accuracy but is not derived from the electroweak Lagrangian; it is an empirical observation whose theoretical status is [\[Open\]](#).

6. Mixing Matrices (CKM and PMNS)

6.1 PMNS (Neutrino Mixing)

All angles are ratios of framework integers. [\[Conjecture\]](#)

$$\boxed{\sin^2 \theta_{12} = \frac{N_c}{N_c + b_3} = \frac{3}{10}, \quad \sin^2 \theta_{23} = \frac{N_{\text{eff}} + N_c}{2N_{\text{eff}} + N_c} = \frac{16}{29}, \quad \sin^2 \theta_{13} = \frac{1}{N_{\text{base}} \cdot N_{\text{eff}}} = \frac{1}{52}} \quad (15)$$

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{(b_3 + N_c)^2}{N_c} = \frac{100}{3} = 33.33 \quad (1.5\% \text{ from } 32.85) \quad \text{[Conjecture]} \quad (16)$$

Normal hierarchy predicted: $m_3 \gg m_2 \gg m_1$, testable by JUNO (~ 2027).

6.2 CKM (Quark Mixing)

Three angles from G^* and framework integers. [\[Conjecture\]](#)

$$\boxed{\sin \theta_C = \frac{G^*}{N_{\text{eff}}}, \quad \sin \theta_{23} = \frac{b_3}{N_{\text{eff}}^2} = \frac{7}{169}, \quad \sin \theta_{13} = \frac{N_c}{b_3 + N_c} \left(\frac{G^*}{N_{\text{eff}}} \right)^3} \quad (17)$$

$$\delta_{CP} = \arctan\left(\frac{b_3}{N_c}\right) = \arctan\left(\frac{7}{3}\right) = 66.8^\circ \quad (2.1\% \text{ from } 65.4^\circ) \quad \text{[Conjecture]} \quad (18)$$

The full CKM matrix via standard PDG parametrization gives all nine $|V_{ij}|$ within 0–8.4% of experiment:

	FTD	PDG 2024	Error
$ V_{ud} $	0.97406	0.97373	0.03%
$ V_{us} $	0.22759	0.22430	1.47%
$ V_{ub} $	0.00354	0.00382	7.4%
$ V_{cd} $	0.22701	0.22100	2.7%
$ V_{cs} $	0.97326	0.97500	0.18%
$ V_{cb} $	0.04142	0.04080	1.5%
$ V_{td} $	0.00867	0.00800	8.4%
$ V_{ts} $	0.04071	0.03880	4.9%
$ V_{tb} $	0.99914	1.01300	1.4%

Jarlskog invariant: $J = c_{12} s_{12} c_{23} s_{23} c_{13}^2 s_{13} \sin \delta = 3.18 \times 10^{-5}$ (3.2% from 3.08×10^{-5}).
First-row unitarity: $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1.000000$ (exact by construction).

7. Precision Formula (Layer 7)

The tree-level $1/\alpha = x_+$ is refined by a four-term correction using the modular deviation $\varepsilon = e^\pi - \pi - (b_3 + N_{\text{eff}}) = e^\pi - \pi - 20 \approx -9.00 \times 10^{-4}$.

$$\boxed{\frac{1}{\alpha} = x_+ - c_1|\varepsilon| + c_2|\varepsilon|^2 - c_3|\varepsilon|^3 - c_4|\varepsilon|^4} \quad (19)$$

where each coefficient is a ratio of framework integers:

$$c_1 = \frac{N_c^2}{\mathcal{D}} = \frac{9}{47}, \quad c_2 = \frac{N_{\text{eff}} - 2N_{\text{base}}}{N_{\text{base}}^3} = \frac{5}{64}, \quad c_3 = \frac{N_{\text{base}}}{N_c \mathcal{D}} = \frac{4}{141}, \quad c_4 = \frac{N_c \mathcal{D}}{b_3 + N_{\text{base}}} = \frac{141}{11} \quad (20)$$

Result: $1/\alpha = 137.035\,999\,177\dots$, matching CODATA 2022 (137.035 999 177(21)) to < 1 part per trillion. **[Conjecture]**

Remark 7.1. The coefficients are pattern-matched, not derived from QED radiative corrections. Four coefficients fitting one target necessarily achieves high precision; the non-trivial content is that each coefficient is expressible as a simple ratio of the same five framework integers $\{3, 4, 7, 13, 47\}$. Falsification: if future CODATA measurements shift outside the correction range.

8. Comparison with Experiment

9. Falsification Criteria

Seven conditions, each sufficient to falsify the framework:

- F1.** Precision α measurement deviating from $x_+ = 137.036\dots$ by > 10 ppm (after accounting for QED corrections).
- F2.** Discovery of a fourth fermion generation with standard gauge couplings.
- F3.** $\sin^2 \theta_W$ measured more than 1% from $3/13$.
- F4.** Definitive determination of inverted neutrino mass hierarchy (JUNO, ~ 2027).
- F5.** Planck-scale Lorentz violation with wrong sign (superluminal high-energy photons).
- F6.** Failure of any integer cascade relation ($b_3 \neq 7$, $N_{\text{eff}} \neq 13$, etc.) under revised QCD data.

F7. Tensor-to-scalar ratio r measured outside $[0.005, 0.036]$ (LiteBIRD, ~ 2032).

10. Reproducibility

All numerical claims are verified by a deterministic proof suite (`scripts/proofs/`, 11 modules, 301 assertions, 301 PASS) requiring only Python 3.10+, NumPy, and SciPy. The master entry point `proof_10_ultimate_chain.py` executes the complete chain. Every constant is computed from scratch in `common.py` with no imports from the project’s own constants module. The computational engine [13] implements these constants as `constexpr` values derived at compile time.

References

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#	Quantity	FTD Formula	FTD	Expt.	Err.	Tag
<i>Coupling constants</i>						
1	$1/\alpha$ (tree)	x_+	137.036	137.036	1.3 ppm	[Conjecture]
2	$1/\alpha$ (corr.)	Eq. (19)	137.036 0	137.036 0	<1 ppt	[Conjecture]
3	$\sin^2\theta_W$	3/13	0.2308	0.2312	0.19%	[Conjecture]
4	$\alpha_s(M_Z)$	7/59	0.1186	0.1179	0.63%	[Conjecture]
5	α_G	Eq. (8)	5.91×10^{-39}	5.91×10^{-39}	0.06%	[Conjecture]
<i>Masses</i>						
6	m_e	Eq. 10	0.5100 MeV	0.51100	0.19%	[Selection]
7	m_μ/m_e	$3b_3(b_3 + N_c) - N_c$	207	206.77	0.11%	[Selection]
8	m_τ/m_e	integer formula	3477	3477.2	0.007%	[Selection]
9	m_p	$N_{\text{eff}} \cdot x_+ + T_{10}$	938.4	938.3	0.02%	[Selection]
10	v (Higgs VEV)	Eq. 11	246.08	246.22	0.05%	[Conjecture]
11	m_H	Eq. 12	125.69	125.25	0.35%	[Selection]
12	m_W	Eq. 13	77.56	80.37	3.5%	[Parametric Insertion]
<i>PMNS mixing</i>						
13	$\sin^2\theta_{12}$	3/10	0.300	0.307	2.3%	[Conjecture]
14	$\sin^2\theta_{23}$	16/29	0.552	0.546	1.0%	[Conjecture]
15	$\sin^2\theta_{13}$	1/52	0.0192	0.0220	12.7% [†]	[Conjecture]
16	Δm^2 ratio	100/3	33.33	32.85	1.5%	[Conjecture]
<i>CKM mixing</i>						
17	$\sin\theta_C$	G^*/N_{eff}	0.2276	0.2243	1.5%	[Conjecture]
18	$\sin\theta_{23}^{CKM}$	7/169	0.0414	0.0408	1.5%	[Conjecture]
19	$\sin\theta_{13}^{CKM}$	$(3/10)(G^*/13)^3$	0.00354	0.00382	7.4%	[Conjecture]
20	δ_{CP}	$\arctan(7/3)$	66.8°	65.4°	2.1%	[Conjecture]
21	Jarlskog J	from angles	3.18×10^{-5}	3.08×10^{-5}	3.2%	[Conjecture]
<i>Cosmology</i>						
22	n_s	$1 - 2/N_e$ ($N_e \approx 56$ e-folds)	0.9645	0.9649	0.04%	[Parametric Insertion]
23	r	$4\alpha(N_c/N_{\text{base}})$	0.0219	< 0.036	OK	[Conjecture]
<i>Structural</i>						
24	N_{gen}	$\lfloor x_- \rfloor$	3	3	exact	[Conjecture]
25	N_c	$\lfloor x_- \rfloor$	3	3	exact	[Conjecture]
26	Hierarchy	normal	NH	NH	—	[Conjecture]

Table 2: FTD predictions vs. experiment. Tags: [Conjecture], [Selection]. Of 26 quantities: 20 within 2%, 24 within 5%, 25 within 13%. [†]Worst prediction; $1/(N_{\text{base}} \cdot N_{\text{eff}})$ is the simplest integer-ratio approximation but overshoots.